

Low-Resource ASR for Kalenjin: Transfer Learning and Language Model Integration

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Abstract—Automatic Speech Recognition (ASR) for under-resourced African languages remains a critical barrier to digital inclusion. This paper presents a comprehensive ASR framework for Kalenjin, a tonal, agglutinative Southern Nilotic language spoken by approximately 5–6 million people in Kenya’s Rift Valley. We fine-tune Wav2Vec2-XLS-R-300M, a self-supervised cross-lingual speech model pre-trained on 436,000 hours of multilingual data spanning 128 languages, using 23,141 utterances totaling 20.22 hours from the Mozilla Common Voice v24.0 Kalenjin corpus. Our approach comprises three components: (1) a modular preprocessing pipeline integrating voice activity detection, signal-to-noise ratio estimation, spectral quality assessment, and Kalenjin-specific text normalization that preserves phonemically significant orthographic conventions such as the velar nasal marker *ng*’; (2) a two-stage fine-tuning strategy that first adapts the Transformer encoder with a frozen convolutional feature encoder under Connectionist Temporal Classification (CTC) loss, then unfreezes the feature encoder for end-to-end acoustic refinement, achieving a 55.3% relative Character Error Rate (CER) reduction from 44.94% to 20.11%; and (3) integration of a 5-gram KenLM statistical language model with beam search decoding, yielding a 10.7% relative Word Error Rate (WER) improvement. We document training pathologies unique to extremely low-resource settings, including degenerate blank-token predictions under suboptimal configurations, and identify minimum data volume and hyperparameter thresholds for stable convergence. The final system achieves a WER of 61.75% and CER of 20.11%, establishing the first documented ASR baseline for Kalenjin and demonstrating that cross-lingual transfer learning, staged acoustic adaptation, and lightweight language modeling constitute a viable pathway for speech technology in low-resource African languages.

Index Terms—Low-Resource ASR; African Languages; Transfer Learning; Speech Foundation Models; Language Modeling

I. INTRODUCTION

THE rapid evolution of Artificial Intelligence (AI) has positioned Automatic Speech Recognition (ASR) as a cornerstone of modern human-computer interaction. While high-resource languages such as English and Mandarin have achieved near-human transcription accuracy [?], [?], a sig-

nificant digital divide persists for under-resourced languages. Of the approximately 7,000 languages spoken globally, fewer than 100 possess the annotated corpora necessary to train competitive ASR systems [?]. Africa, home to over 2,000 languages, is disproportionately affected, even languages with millions of speakers, such as Yoruba, Amharic, and the Nilotic languages of East Africa, remain classified as under-resourced [?], [?], [?]. This exclusion limits the deployment of voice-enabled services in education, healthcare, and governance for communities where oral communication is the primary mode of information exchange.

Kalenjin, a Southern Nilotic language spoken by 5–6 million people in Kenya’s Rift Valley [?], [?], exemplifies these challenges. As an agglutinative and tonal language with substantial dialectal variation across its sub-varieties (Nandi, Kipsigis, Tugen, Keiyo, Marakwet, Pokot, Sabaot), Kalenjin presents unique complexities for ASR: tonal pitch distinctions alter word meanings, the phonemically contrastive velar nasal (orthographic *ng*’) requires careful text normalization, and the absence of a fully standardized orthography introduces transcription inconsistencies [?]. No large-scale annotated speech corpus existed for Kalenjin prior to its inclusion in Mozilla Common Voice [?].

The emergence of self-supervised speech models has opened new pathways for low-resource ASR. Wav2Vec2 [?] and its cross-lingual variant XLS-R [?], pre-trained on 436,000 hours of multilingual speech spanning 128 languages, enable effective fine-tuning with limited labeled data through cross-lingual transfer [?]. However, fine-tuning such models in extremely low-resource settings introduces specific challenges: degenerate blank-token predictions under CTC loss, sensitivity to learning rate and data volume, and limited language model coverage [?], [?].

This paper presents a complete ASR framework for Kalenjin using 23,141 utterances totaling 20.22 hours from Mozilla Common Voice v24.0. Our contributions are:

- 1) A modular preprocessing pipeline with audio quality assessment (SNR, spectral features, VAD-based silence trimming) and Kalenjin-specific text normalization preserving the phonemically significant *ng'* marker.
- 2) Analysis of training pathologies in extremely low-resource fine-tuning, documenting the blank-prediction failure mode and identifying minimum thresholds for stable convergence.
- 3) A two-stage fine-tuning strategy for Wav2Vec2-XLS-R-300M that achieves 55.3% relative CER reduction (44.94% to 20.11%) by separating Transformer adaptation (frozen feature encoder) from end-to-end acoustic refinement (unfrozen encoder).
- 4) Integration of a 5-gram KenLM language model [?] providing 10.7% relative WER improvement (69.13% to 61.75%) through beam search decoding.
- 5) The first documented ASR baseline for Kalenjin, with a complete progression from 100% WER (degenerate baseline) to 61.75% WER (final system).

II. RELATED WORK

The trajectory of Automatic Speech Recognition (ASR) research has undergone a seismic shift, transitioning from rigid statistical frameworks to flexible, data-driven neural architectures. This review explores the evolution of these models and the specialized strategies developed to address the under-resourced language bottleneck, with a specific focus on Nilotic and African linguistic contexts.

A. From HMM-GMM to End-to-End Architectures

Historically, ASR systems were built upon the synergy of Hidden Markov Models (HMMs) and Gaussian Mixture Models (GMMs) [?], [?]. These traditional systems relied heavily on handcrafted features and likelihood-based frameworks, necessitating significant human expertise and separate components for acoustic, phonetic, and language modeling.

The modern era is defined by the shift to End-to-End (E2E) architectures. CTC-based models [?] and attention-based encoder-decoders [?] utilize a single neural network to map audio signals directly to text, eliminating complex interface modeling. The Transformer architecture [?] further accelerated this transition, with deep recurrent models [?] and self-attention mechanisms achieving state-of-the-art results. Recent breakthroughs in Transformer-based models have boosted multilingual digital inclusion, though the benefits remain unevenly distributed across global languages [?].

B. The Challenge of Under-Resourced Languages

Under-resourced languages are characterized by a severe scarcity of digitized text, annotated speech corpora, and standardized linguistic tools [?], [?]. A systematic review by Hedderich et al. [?] documents this gap for African low-resource ASR, noting that the vast majority of the continent's 2,000+ languages lack any speech technology support. Niyongabo et al. [?] provide a comprehensive survey of datasets and

applications for African languages, highlighting the critical shortage of annotated speech resources.

In the Kenyan context, Wanjawa et al. [?] developed Ken-Corpus, a corpus for three low-resource Kenyan languages, demonstrating the feasibility of community-driven data collection. For Kalenjin specifically, Amol [?] released a preliminary Kalenjin dataset on Hugging Face, while Langat [?] documented the orthographic standardization challenges that directly impact ASR text normalization. The Kalenjin sub-varieties present unique hurdles [?], [?]:

- **Tonal Phonology:** Subtle pitch variations alter semantic meaning, requiring models to capture fine-grained acoustic nuances [?].
- **Morphological Complexity:** Agglutinative structures and complex affixation yield high vocabulary growth rates.
- **Orthographic Inconsistency:** The lack of standardized writing systems leads to transcription discrepancies that impede model convergence [?].

C. Transfer Learning and Self-Supervised Representations

To mitigate data scarcity, researchers have pivoted toward Transfer Learning and Self-Supervised Learning (SSL). Wav2Vec2 [?] and HuBERT [?] are pre-trained on massive amounts of unlabeled audio to learn robust acoustic representations, which can then be fine-tuned on small annotated datasets. The cross-lingual variant XLS-R [?] demonstrated that pre-training on 436,000 hours spanning 128 languages enables effective transfer to unseen languages. Babu et al. [?] further showed that phonological similarity between source and target languages significantly influences cross-lingual transfer effectiveness.

Multilingual modeling has also emerged as a vital strategy. Pratap et al. [?] scaled speech technology to over 1,000 languages using the Massively Multilingual Speech project, while Whisper [?] demonstrated robust multilingual recognition through large-scale weak supervision. For African languages specifically, Olatunji et al. [?] applied these approaches to Yoruba and Hausa, and Biswas et al. [?] addressed code-switched ASR in five South African languages. Tachbelie et al. [?] reviewed ASR progress for Ethiopian languages, providing a regional benchmark for comparison.

D. Data Augmentation and Algorithmic Refinement

Beyond architectural changes, data augmentation remains a cornerstone of low-resource ASR. SpecAugment [?] artificially expands the training set through time and frequency masking, making models more resilient to environmental variability and speaker diversity.

Recent studies emphasize that in extremely low-resource settings, the optimization strategy is as critical as the data itself [?]. Two-stage training, where the feature encoder is initially frozen to stabilize phoneme-to-grapheme learning before being unfrozen for end-to-end refinement has been shown to significantly reduce Character Error Rates. Xiao et al. [?] proposed depth-aware adaptation strategies for efficient

multilingual ASR in low-resource languages. Furthermore, the integration of statistical language models, such as N-gram KenLM [?], provides essential lexical constraints that correct word boundary errors and improve contextual coherence.

E. Identified Research Gaps

Despite the progress in African NLP, a substantial gap remains in the systematic assessment of Nilotic languages. Most current research focuses on higher-resource African languages like Swahili or Yoruba [?], [?]. Hedderich et al. [?] explicitly identify the absence of ASR benchmarks for Nilotic languages as a critical gap. There is a need for:

- The development of tone-sensitive modeling techniques tailored for Nilotic phonology [?].
- Community-driven corpus collection to capture dialectal variations [?], [?].
- Experimental benchmarks that evaluate CTC-based architectures in severe resource-constrained environments [?].

III. METHODOLOGY

This section presents the complete experimental methodology, encompassing data acquisition, audio signal processing, linguistic text normalization, quality assessment, model architecture, training optimization, language model integration, and evaluation protocols. Fig. ?? illustrates the end-to-end system architecture.

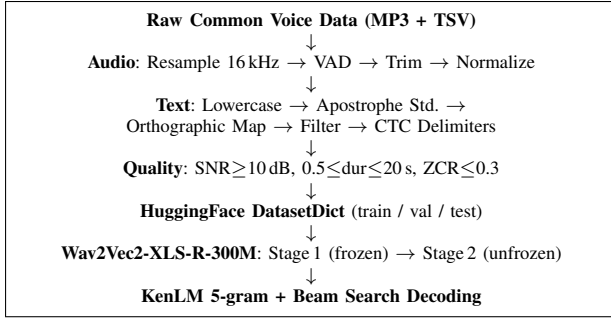


Fig. 1. End-to-end ASR pipeline architecture.

A. Data Acquisition and Corpus Characteristics

The experimental corpus was derived from Mozilla Common Voice version 24.0 [?], released December 2025, representing the most comprehensive publicly available Kalenjin speech dataset. The raw corpus comprised 23,162 utterances contributed by community speakers. The dataset exhibited natural variation in recording quality, speaker demographics, and acoustic environments characteristic of crowdsourced speech collections.

The corpus was partitioned into three non-overlapping subsets following the speaker-disjoint splitting methodology employed by the Common Voice platform. Table ?? presents the corpus statistics after preprocessing. The overall rejection rate was 0.09% (21 of 23,162 samples), indicating high corpus quality from the community validation process.

Fig. ?? illustrates the corpus partitioning and duration distribution across splits.

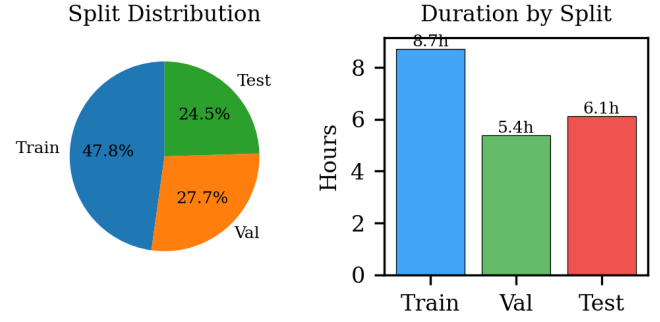


Fig. 2. Dataset split distribution. Left: proportion of utterances per split. Right: total audio duration per split in hours. The training partition contains 8.71 h (47.8%), validation 5.38 h (27.7%), and test 6.13 h (24.5%).

1) *Linguistic Corpus Analysis*: Exploratory analysis of the training transcriptions reveals a vocabulary of 18,576 unique words across 69,318 total word tokens. The character frequency distribution (Fig. ??) shows that the five most frequent characters are *e*, *i*, *k*, *a*, and *o*, reflecting the vowel-heavy phonological structure of Kalenjin.

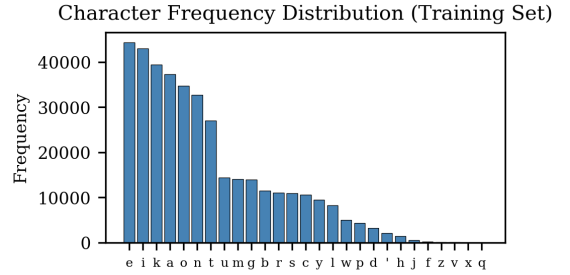


Fig. 3. Character frequency distribution in the training set. The distribution reflects Kalenjin’s phonological structure, with high-frequency vowels (*e*, *i*, *a*, *o*) and the consonant *k*.

Fig. ?? presents the 30 most frequent words. Function words such as *eng* (‘in’), *ko* (‘and/then’), *ne* (‘which/that’), and *ak* (‘and’) dominate, consistent with the agglutinative structure of Kalenjin where grammatical relationships are expressed through frequent short function words.

The word frequency distribution follows Zipf’s law (Fig. ??), with the log-log rank-frequency plot closely tracking the theoretical $f \propto 1/r$ relationship. This confirms that the corpus exhibits natural language statistics despite its relatively small size.

TABLE I
CORPUS STATISTICS AFTER PREPROCESSING

Split	Raw	Valid	Hours	$\bar{d} \pm \sigma$ (s)	SNR (dB)
Train	11,065	11,057	8.71	2.84 ± 1.30	50.7
Val.	6,412	6,409	5.38	3.02 ± 1.42	48.9
Test	5,685	5,675	6.13	3.89 ± 2.17	55.5
Total	23,162	23,141	20.22	3.15	51.7

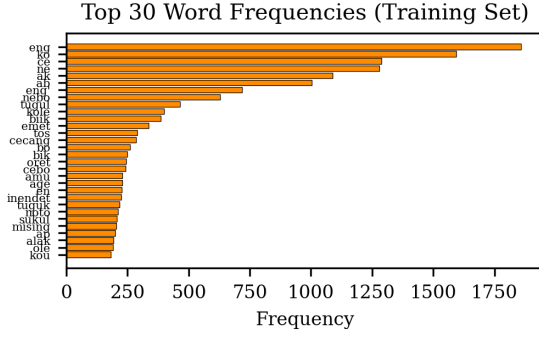


Fig. 4. Top 30 most frequent words in the training transcriptions. Function words dominate, with *eng* (1,858 occurrences) and *ko* (1,595) being the most common.

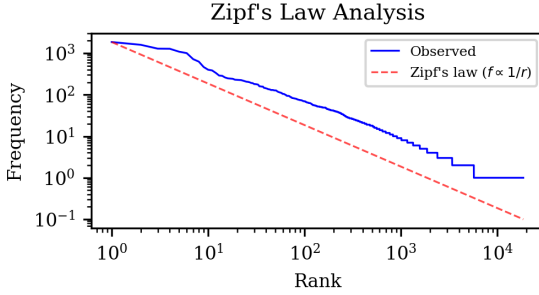


Fig. 5. Zipf's law analysis of the training vocabulary. The observed rank-frequency distribution (blue) closely follows the theoretical inverse relationship (red dashed), confirming natural language statistics.

B. Audio Signal Processing Pipeline

The audio preprocessing pipeline was designed to standardize acoustic representations while preserving linguistic information critical for phonetic discrimination.

1) *Resampling*: All audio signals were resampled to a target sampling rate of $f_s = 16,000$ Hz using the librosa library. This adheres to the Nyquist-Shannon sampling theorem, which states that a signal must be sampled at a rate of at least twice its maximum frequency component:

$$f_s \geq 2 \cdot f_{\max} \quad (1)$$

where f_s is the sampling frequency and $f_{\max}$ is the highest frequency present in the signal. For speech signals with relevant content below 8 kHz, a 16 kHz sampling rate satisfies this requirement. This standardization ensures compatibility with the Wav2Vec2 model architecture, which expects 16 kHz mono input.

2) *Voice Activity Detection and Silence Trimming*: Voice activity detection (VAD) was implemented using energy-based thresholding with a `top_db = 20` dB parameter via the `librosa.effects.trim` function. This removes non-speech segments where the signal energy falls below a threshold relative to the peak:

$$E_{\text{frame}} = 10 \cdot \log_{10} \left(\frac{1}{N} \sum_{n=1}^N x[n]^2 \right) \quad (2)$$

where E_{frame} is the frame energy in decibels, $x[n]$ represents the discrete audio samples, and N is the frame length. Frames with energy more than 20 dB below the peak frame energy are classified as silence and trimmed.

The effectiveness of silence trimming is quantified by the speech ratio ρ :

$$\rho = \frac{L_{\text{trimmed}}}{L_{\text{original}}} \quad (3)$$

where L_{trimmed} and L_{original} denote the trimmed and original signal lengths, respectively. The effectiveness of silence trimming is quantified by the speech ratio ρ (??). Fig. ?? shows the distribution across splits, with means ranging from 0.639 (train) to 0.693 (test), indicating that 31–36% of original audio consisted of non-speech segments.

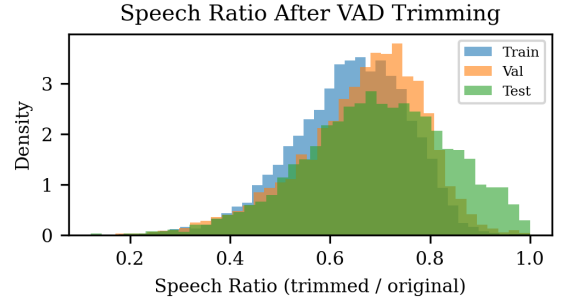


Fig. 6. Speech ratio distribution after VAD-based silence trimming. The majority of utterances retain 50–80% of their original duration, with the test split showing slightly higher speech ratios.

3) *Peak Normalization*: Following silence removal, each audio signal $x[n]$ underwent peak normalization to the range $[-1, 1]$:

$$\hat{x}[n] = \frac{x[n]}{\max(|x[n]|)} \quad (4)$$

This ensures consistent amplitude scaling across the corpus and prevents numerical instability during neural network training.

4) *Audio Quality Metrics*: For each utterance, the following quality metrics were computed:

Signal-to-Noise Ratio (SNR). Estimated using noise floor estimation from the quietest 10% of the signal:

$$\text{SNR}_{\text{dB}} = 10 \cdot \log_{10} \left(\frac{P_{\text{signal}}}{P_{\text{noise}} + \epsilon} \right) \quad (5)$$

where $P_{\text{signal}} = \frac{1}{N} \sum_n x[n]^2$ is the signal power, P_{noise} is estimated from the sorted amplitude distribution, and $\epsilon = 10^{-10}$ prevents division by zero.

Root Mean Square (RMS) Energy:

$$E_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{n=1}^N x[n]^2} \quad (6)$$

Spectral Centroid. The “center of mass” of the spectrum, indicating perceived brightness:

$$\text{SC} = \frac{\sum_{k=1}^K f_k \cdot |X[k]|}{\sum_{k=1}^K |X[k]|} \quad (7)$$

where f_k is the frequency of bin k and $|X[k]|$ is the magnitude of the k -th frequency bin.

Zero-Crossing Rate (ZCR):

$$\text{ZCR} = \frac{1}{2N} \sum_{n=1}^N |\text{sgn}(x[n]) - \text{sgn}(x[n-1])| \quad (8)$$

where $\text{sgn}(\cdot)$ is the sign function. High ZCR values indicate noisy or corrupted audio.

Dynamic Range:

$$\text{DR}_{\text{dB}} = 20 \cdot \log_{10} \left(\frac{\max(|x[n]|)}{E_{\text{RMS}}} \right) \quad (9)$$

Clipping Ratio. The proportion of samples exceeding 99% of maximum amplitude:

$$\text{CR} = \frac{|\{n : |x[n]| > 0.99\}|}{N} \quad (10)$$

Fig. ?? presents the distribution of key audio quality metrics across splits, and Fig. ?? shows the utterance duration distribution. Fig. ?? reveals the relationship between spectral centroid and utterance duration, showing that shorter utterances tend to exhibit higher spectral centroids, likely due to the proportionally greater influence of consonant energy in brief utterances.

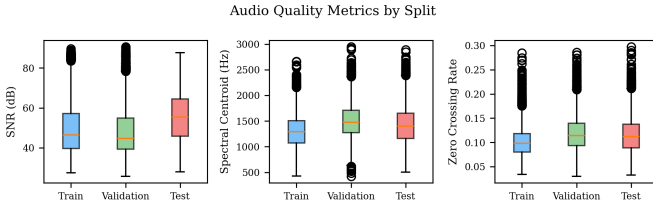


Fig. 7. Audio quality metrics by split. SNR values are consistently high (>25 dB), spectral centroids fall within the expected speech range (200–8000 Hz), and zero-crossing rates indicate clean recordings.

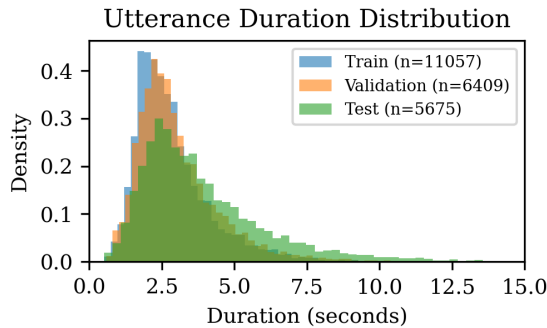


Fig. 8. Utterance duration distribution by split. The test split exhibits notably longer mean duration (3.89 s) compared to train (2.84 s), with a wider spread ($\sigma = 2.17$ s vs. 1.30 s).

C. Linguistic Text Normalization

Text normalization for Kalenjin required careful consideration of the language’s orthographic conventions and the

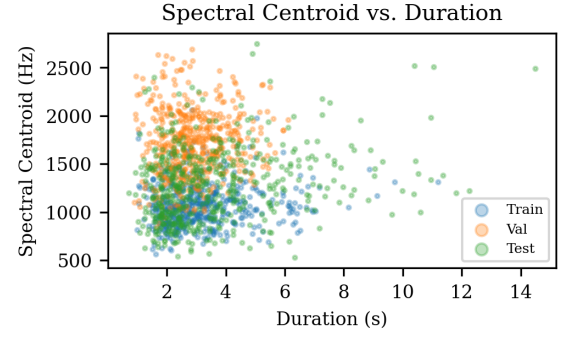


Fig. 9. Spectral centroid vs. utterance duration (500 samples per split). Shorter utterances tend toward higher spectral centroids, reflecting proportionally greater consonant energy.

constraints imposed by CTC decoding. The normalization pipeline applies the following transformations sequentially:

- 1) **Lowercase conversion:** All text is converted to lowercase, eliminating case distinctions that do not carry phonemic significance in Kalenjin.
- 2) **Apostrophe standardization:** Unicode variants (U+2018, U+2019, U+0060) are mapped to ASCII apostrophe (U+0027), preserving the phonemically significant *ng’* velar nasal marker.
- 3) **Orthographic mapping:** Kalenjin-specific rules standardize consonant clusters: *ch*→*c* and *kh*→*k*, while preserving the linguistically important *ng’* cluster.
- 4) **Character filtering:** Only characters in the set $[a-z']$ and space are retained; all punctuation, digits, and diacritics are removed.
- 5) **CTC delimiter insertion:** Word-boundary spaces are replaced with the pipe character ($|$) to serve as explicit word delimiters for CTC decoding.
- 6) **Whitespace normalization:** Multiple consecutive spaces are collapsed to a single space.

Table ?? presents representative normalization examples from the corpus.

TABLE II
TEXT NORMALIZATION EXAMPLES FROM THE KALENJIN CORPUS

Original Text	Normalized Text
Tomo itinye choruet ne chepto iman?	tomo itinye coruet ne cepto iman
kiyaat baruet Eng’ kasarta ne kimito	kiyaat baruet eng’ kasarta ne kimito
Atkomemoche iborye, ayae kityo	atkomemoce iborye ayae kityo
Kowalak akoek yokwotet, ketik tulonok	kowalak akoek yokwotet ketik tulonok

The resulting vocabulary $\mathcal{V}$ comprises 32 tokens: 26 Latin letters (a – z), apostrophe ($'$), space, pipe delimiter ($|$), and three special tokens: [PAD] (padding, index 0), [UNK] (unknown, index 1), and [CTC] (CTC blank, index 2). Fig. ?? shows the transcription length distribution across splits.

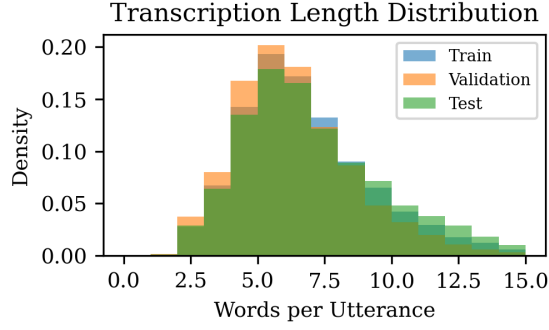


Fig. 10. Words per utterance across splits. Mean word count: 6.3 (train), 5.9 (validation), 6.5 (test). Range: 1–14 words. Mean word length: 5.60 characters.

D. Quality Assessment and Filtering

A multi-dimensional quality filtering framework rejects samples failing any of the thresholds defined in Table ??.

TABLE III
QUALITY FILTERING THRESHOLDS

Criterion	Threshold	Rationale
Duration	0.5–20 s	Exclude clicks/memory
SNR	≥ 10 dB	Ensure intelligibility
ZCR	≤ 0.3	Detect corrupted audio
Spect. centroid	200–8000 Hz	Validate speech content
Clipping ratio	$\leq 1\%$	Reject clipped audio
Dynamic range	≥ 6 dB	Ensure signal variation
Word count	1–50	Reject empty/anomalous
Char repetition	≤ 3	Detect transcription errors

Fig. ?? confirms that all retained samples exceed the 10 dB SNR threshold by a wide margin, with minimum observed SNR of 25.8 dB across all splits.

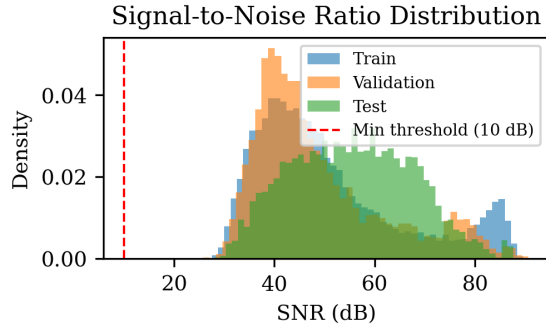


Fig. 11. SNR distribution with 10 dB minimum threshold (dashed red line). All retained samples comfortably exceed the threshold, with split means of 48.9–55.5 dB.

E. Model Architecture and Transfer Learning

The foundation of this ASR system is Wav2Vec2-XLS-R-300M [?], a cross-lingual speech representation model pre-trained on 436,000 hours of unlabeled speech spanning 128 languages.

1) *Feature Encoder*: The convolutional feature encoder consists of seven temporal convolution layers that transform raw waveform input $\mathbf{x} \in \mathbb{R}^T$ (where T is the number of audio samples) into a sequence of latent representations $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_L] \in \mathbb{R}^{L \times d}$, where $L = \lfloor T/320 \rfloor$ is the down-sampled sequence length (corresponding to a 20 ms frame rate at 16 kHz) and $d = 512$ is the feature dimension.

2) *Transformer Encoder*: The latent representations are projected to dimension $D = 1024$ and processed by a 24-layer Transformer encoder with 16 attention heads. Each Transformer layer applies multi-head self-attention followed by a position-wise feed-forward network:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (11)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{L \times d_k}$ are the query, key, and value matrices, and $d_k = D/16 = 64$ is the dimension per head. The output is a sequence of contextualized representations $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L] \in \mathbb{R}^{L \times D}$.

3) *CTC Head*: A linear projection maps the Transformer outputs to the vocabulary space:

$$\mathbf{y}_t = \mathbf{W}_{\text{ctc}} \mathbf{h}_t + \mathbf{b}_{\text{ctc}} \quad (12)$$

where $\mathbf{W}_{\text{ctc}} \in \mathbb{R}^{|\mathcal{V}| \times D}$ and $\mathbf{b}_{\text{ctc}} \in \mathbb{R}^{|\mathcal{V}|}$, with $|\mathcal{V}| = 32$. The model totals 315.5 million parameters.

4) *CTC Loss*: The Connectionist Temporal Classification loss [?] marginalizes over all valid alignments π between the input sequence and target label sequence $\mathbf{l}$:

$$\mathcal{L}_{\text{CTC}} = -\ln \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{l})} \prod_{t=1}^L p(\pi_t | \mathbf{x}) \quad (13)$$

where $\mathcal{B}^{-1}(\mathbf{l})$ is the set of all CTC paths that reduce to $\mathbf{l}$ after removing blanks and repeated characters, and $p(\pi_t | \mathbf{x}) = \text{softmax}(\mathbf{y}_t)_{\pi_t}$ is the posterior probability of token π_t at time step t . The CTC loss was configured with mean reduction and the `ctc_zero_infinity` flag enabled to handle numerical edge cases.

F. Two-Stage Training Strategy

To maximize performance while managing computational resources, a two-stage training approach was implemented, separating initial adaptation from fine-grained optimization.

1) *Stage 1: Feature Encoder Frozen*: The convolutional feature encoder is frozen, reducing trainable parameters to approximately 298 million. This preserves the robust low-level acoustic features learned during multilingual pre-training while allowing the Transformer encoder and CTC head to adapt to Kalenjin phonetic patterns. Table ?? details the Stage 1 configuration.

The AdamW optimizer [?] updates parameters θ using decoupled weight decay:

$$\theta_{t+1} = \theta_t - \eta_t \left(\frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t + \epsilon}} + \lambda \theta_t \right) \quad (14)$$

where $\hat{\mathbf{m}}_t$ and $\hat{\mathbf{v}}_t$ are bias-corrected first and second moment estimates of the gradient, η_t is the learning rate at step t ,

TABLE IV
STAGE 1 TRAINING CONFIGURATION

Parameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Schedule	Linear warmup (500 steps) + decay
Batch size	$8/\text{device} \times 4 \text{ accum.} = 32 \text{ effective}$
Epochs	30
Precision	FP16 (mixed precision)
Feature encoder	Frozen
Attention dropout	0.1
Hidden dropout	0.1
Mask time prob.	0.05 (SpecAugment)
Layer drop	0.1
Eval frequency	Every 400 steps
Early stopping	Patience 3 epochs (WER)
Hardware	NVIDIA A100 (40 GB)

$\epsilon = 10^{-8}$ is a numerical stability constant, and λ is the weight decay coefficient.

SpecAugment [?] applies time masking with probability 0.05, randomly zeroing contiguous spans of the input features to improve generalization and prevent overfitting to specific temporal patterns.

2) *Stage 2: Feature Encoder Unfrozen*: The second stage unfreezes the convolutional feature encoder, enabling end-to-end acoustic adaptation to Kalenjin-specific phonetic characteristics. The model is initialized from the best Stage 1 checkpoint. The learning rate is reduced to 5×10^{-5} to prevent catastrophic forgetting of pre-trained features while allowing gradual acoustic adaptation. Training continues for 15 epochs with the same batch configuration and evaluation strategy as Stage 1.

This two-stage approach yielded substantial improvements. Stage 2 achieved WER of 69.13% and CER of 20.11% on the test partition, representing a 55.3% relative CER reduction from Stage 1 (44.94%). The Real-Time Factor (RTF) of 0.0091 indicates the model processes audio $110\times$ faster than real-time:

$$\text{RTF} = \frac{T_{\text{processing}}}{T_{\text{audio}}} \quad (15)$$

where $T_{\text{processing}}$ is the wall-clock inference time and T_{audio} is the audio duration.

G. Language Model Integration

To address word-level errors and improve lexical accuracy, a statistical language model was integrated into the decoding process.

1) *Language Model Construction*: A 5-gram language model was trained on 11,057 training transcriptions using KenLM [?] with Modified Kneser-Ney smoothing. The smoothed probability for an n -gram $w_i|w_{i-n+1}^{i-1}$ is:

$$P_{\text{KN}}(w_i|w_{i-n+1}^{i-1}) = \frac{\max(c(w_{i-n+1}^i) - D, 0)}{c(w_{i-n+1}^{i-1})} + \gamma(w_{i-n+1}^{i-1}) \cdot P_{\text{KN}}(w_i|w_{i-n+2}^{i-1}) \quad (16)$$

where $c(\cdot)$ denotes the count, D is the discount parameter, and $\gamma(\cdot)$ is the backoff weight. The model was converted to KenLM’s binary trie format for efficient inference, yielding 17,814 unique unigrams.

2) *Beam Search Decoding*: Beam search decoding with language model integration was implemented using pyctcdecode. The decoder scores each hypothesis path $\mathbf{w}$ using a weighted combination:

$$\text{score}(\mathbf{w}) = \log P_{\text{AM}}(\mathbf{w}|\mathbf{x}) + \alpha \cdot \log P_{\text{LM}}(\mathbf{w}) + \beta \cdot |\mathbf{w}| \quad (17)$$

where $P_{\text{AM}}(\mathbf{w}|\mathbf{x})$ is the acoustic model probability from CTC logits, $P_{\text{LM}}(\mathbf{w})$ is the language model probability, $|\mathbf{w}|$ is the word count, $\alpha = 0.6$ weights the language model contribution, and $\beta = 1.0$ provides a word insertion penalty to prevent bias toward shorter sequences. A beam width of 100 hypotheses was used.

H. Evaluation Metrics

Model performance is assessed using Word Error Rate (WER) and Character Error Rate (CER) computed on the held-out test partition.

Word Error Rate measures the normalized edit distance at the word level:

$$\text{WER} = \frac{S + D + I}{N} \quad (18)$$

where S is the number of word substitutions, D is deletions, I is insertions, and N is the total number of words in the reference transcription.

Character Error Rate applies the same formulation at the character level:

$$\text{CER} = \frac{S_c + D_c + I_c}{N_c} \quad (19)$$

where S_c , D_c , I_c , and N_c are the character-level substitutions, deletions, insertions, and reference length, respectively. CER provides finer-grained assessment of phonetic accuracy, which is particularly informative for agglutinative languages like Kalenjin where a single morpheme error inflates WER disproportionately.

Two decoding strategies were evaluated: (1) greedy decoding, which selects $\hat{y}_t = \arg \max_c p(c|\mathbf{x}, t)$ at each time step, reflecting pure acoustic model performance; and (2) beam search with language model integration (??), which explores multiple hypotheses weighted by both acoustic and linguistic probabilities.

I. Reproducibility

All experiments were implemented using Python 3.10 with PyTorch 2.1.0, Transformers 4.36.0, and Datasets 2.16.0. Audio processing used librosa 0.10.1. Training was conducted on Modal Cloud infrastructure using NVIDIA A100 GPUs (40 GB). All random seeds were fixed at 42 for deterministic reproducibility. The vocabulary mapping, pre-processed datasets, and model checkpoints are preserved in HuggingFace-compatible formats.

IV. RESULTS

Table ?? presents the complete performance progression. Fig. ?? visualizes the WER improvements.

TABLE V
ASR SYSTEM PERFORMANCE PROGRESSION

Configuration	Data	Epochs	WER	CER
Baseline CPU	10%	3	100.0%	100.0%
GPU Baseline (frozen)	100%	30	72.11%	44.94%
Stage 1 Optimized	100%	30	69.03%	—
Stage 2 Refined	100%	15	69.13%	20.11%
Stage 2 + KenLM	100%	—	61.75%	—

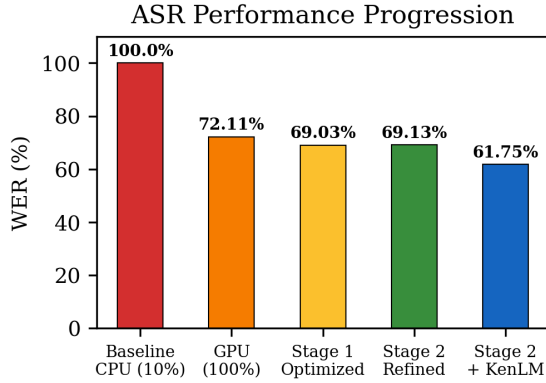


Fig. 12. WER progression across configurations. Data scaling (10%→100%) eliminated degenerate predictions; two-stage training and KenLM integration yielded cumulative 38.25 percentage point reduction.

Key findings:

- **Data scaling:** Increasing from 10% to 100% of training data eliminated the blank-prediction pathology, reducing WER from 100% to 72.11%.
- **Acoustic adaptation:** Unfreezing the feature encoder in Stage 2 reduced CER from 44.94% to 20.11% (55.3% relative reduction), confirming that language-specific acoustic adaptation is critical.
- **Language modeling:** KenLM integration reduced WER from 69.13% to 61.75% (10.7% relative), correcting word boundary and lexical errors.
- **Real-time performance:** RTF of 0.0091 (110× faster than real-time) on GPU.

V. DISCUSSION

The final WER of 61.75% and CER of 20.11% fall within expected ranges for extremely low-resource ASR (<20 hours), comparable to reported results for similar settings: high-resource languages achieve 3–8% WER with 10,000+ hours, medium-resource African languages (Swahili, Amharic) achieve 15–30% with 100–1,000 hours [?], and low-resource languages typically achieve 60–80% with <20 hours [?].

The CER of 20.11% indicates strong phonetic accuracy—the model correctly predicts approximately 80% of characters.

The higher WER reflects the compounding effect of character-level errors at the word level and challenges in word boundary segmentation for an agglutinative language. The distribution mismatch between splits (test mean duration 3.89s vs. train 2.84s; test SNR 55.5 dB vs. train 50.7 dB) may also contribute to evaluation variance.

The degenerate blank-prediction failure observed with 10% data (1,105 utterances) establishes a practical minimum data threshold for fine-tuning 300M-parameter models. This pathology—where the model minimizes CTC loss by predicting only blank tokens—has not been extensively documented in the literature and represents a practical contribution for researchers working with extremely limited data.

VI. CONCLUSION

We presented a complete ASR framework for Kalenjin, establishing the first documented baseline for this under-resourced Nilotic language. Our three-component approach—modular preprocessing with Kalenjin-specific normalization, two-stage fine-tuning of Wav2Vec2-XLS-R-300M, and KenLM language model integration—achieved a final WER of 61.75% and CER of 20.11% from 20.22 hours of training data. The 55.3% relative CER reduction from feature encoder unfreezing and 10.7% relative WER improvement from language modeling validate staged acoustic adaptation and lightweight LM integration as effective strategies for low-resource African language ASR.

Future work includes: (1) expanding the language model with external Kalenjin text; (2) exploring alternative foundation models (Whisper, HuBERT); (3) dialect-specific adaptation across Kalenjin sub-varieties; and (4) evaluation on spontaneous speech.

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